FLEXBMS

System Overview

Home battery energy storage system

Document date	12 September 2026
System	96s LiFePO4 home BESS
Nominal capacity	314 Ah / approximately 96.5 kWh
Primary controller	STM32G491
Document status	Living architecture overview

This document explains the current FlexBMS system at architecture level. Schematics and source code remain authoritative for implementation details.

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1. System level

FlexBMS is the battery management and integration platform for a stationary home battery energy storage system. The battery consists of 96 LiFePO4 cells in series, arranged as four 24-cell modules. At a nominal cell voltage of 3.2 V and a capacity of 314 Ah, the pack is approximately 307 V and 96.5 kWh.

The system connects the battery to a three-phase hybrid inverter. FlexBMS measures the battery, protects it, controls the high-voltage connection, and provides the information needed by higher-level energy management. The battery management system remains the final authority for battery safety.

1.1. System roles

Layer	Responsibility	Status
Battery and HV hardware	Stores energy, measures physical quantities, and safely connects the pack to the inverter.	CURRENT
STM32 BMS	Performs real-time measurement, protection, state handling, and contactor control.	CURRENT
ESP32 gateway	Provides local BMS maintenance, Wi-Fi provisioning, and telemetry transport while remaining listen-only on CAN.	CURRENT
Energy management system	Optimizes charge and discharge requests within BMS, inverter, and site limits. Shadow scheduling is operational; automatic actuation is not yet enabled.	SHADOW TEST
Home Assistant	Provides dashboards, automation, and a user-facing view of the installation.	CURRENT

1.2. Energy and information flow

Grid, PV and home	<->	**Hybrid inverter**	<->	**FlexBMS battery**
Home Assistant	<->	EMS	->	Operating request

The BMS may reduce or reject any request to protect the battery.

Control principle

Energy management is advisory; battery protection is mandatory. The EMS may request an operating point, but the BMS defines the permitted charge and discharge envelope and can open the contactor.

2. Hardware

2.1. Battery assembly

The pack contains 96 series-connected LiFePO4 cells divided into four physical 24s modules. Each module contains two twelve-cell monitoring sections. A 200 A / 75 mV shunt is located at the negative end of the pack for current measurement.

Item	Current baseline
Cell chemistry	LiFePO4
Series configuration	96s
Physical modules	4 x 24s
Cell capacity	314 Ah
Nominal pack voltage	307.2 V
Nominal stored energy	Approximately 96.5 kWh
Current shunt	200 A / 75 mV

The modules are installed in an unheated garage. Temperature measurement is distributed across the pack. The current STM32 configuration expects four NTC inputs per battery-monitor IC. Final sensor population and harness documentation should be kept consistent with the assembled modules.

2.2. Module boards

Each 24s module has one FlexBMS module board with two MC33771C battery cell controllers. Each controller measures twelve cells, supports passive balancing, reads local temperature sensors, and participates in the transformer-isolated TPL daisy chain.

Across the complete pack this gives:

- eight MC33771C devices;
- 96 monitored cell voltages;
- up to 32 configured NTC measurements; and
- one current-sensing location on the first, lowest-voltage controller.

2.3. Master controller

The master board is the central interface between the battery, high-voltage switching hardware, service tools, inverter side, and local network.

Its main functional blocks are:

- STM32G491 real-time controller;
- MC33664 TPL transceiver for the module-board chain;
- ESP32-C3-WROOM-02U connectivity controller;
- isolated CAN and RS485 interfaces;
- isolated measurements of battery-side and load-side high voltage;
- contactor and precharge relay drivers;
- USB for flashing, diagnostics, and the Companion application; and
- power selection between USB and the battery-derived low-voltage supply.

The board contains separate HV, LV, and inverter-side isolated domains. USB-only power supports development and diagnostics, but does not provide the 12 V rail required to energize the contactor.

2.4. High-voltage path

The main contactor is located in the positive battery path. A separate precharge path limits inrush current before the contactor closes. The STM32 is the sole software owner of both drivers. It measures the battery and load sides through separate isolated AMC3330 channels, compares the battery-side result with the voltage reported by the cell-monitor chain, and observes both active-low diagnostic outputs.

The contactor is pulled in at full drive and then held with 20 kHz PWM. Precharge remains closed during pull-in and opens only after the contactor drive is established. The hold duty is a firmware configuration value.

Voltage feedback shows whether the two HV nodes are plausible and whether precharge has progressed. It does not prove the mechanical position of the contactor: the load-side voltage may remain present after opening because the inverter DC-link discharge behavior is not yet characterized.

Pack-level protection also includes the main DC fuse and a manual DC isolation device. Exact component ratings, wiring, creepage, clearance, and protection coordination belong in the hardware design files and commissioning documentation.

2.5. Physical communications

Interface	Connection	Purpose
TPL	Master to 8 monitor ICs	Cell, temperature, diagnostic, and current data.
USB	Service computer to STM32	Firmware, logs, and Companion access.
CAN	STM32 and inverter-side domain	Battery status and inverter communication.
Wi-Fi	ESP32 to local network	Local maintenance page, Wi-Fi provisioning, and future telemetry integration.
RS485	ESP32 to external energy meters	Optional read-only Modbus RTU polling of up to two SDM72D-M-2 meters.

3. Software

The software is divided by responsibility. Time-critical protection and actuation remain on the STM32. Network-facing functions belong on the ESP32. Site optimization belongs in the EMS, with Home Assistant providing the local user and automation layer.

3.1. STM32

CURRENT

The STM32 firmware is the real-time BMS. It starts and monitors the MC33771C chain, gathers a complete battery measurement set, evaluates battery and communication faults, and controls relay permission.

The current battery profile is:

- eight MC33771C devices with twelve cells each;
- four NTC channels per controller on GPIO0 through GPIO3;
- current sensing on CID 1 with a 375 micro-ohm shunt value;
- positive current for charging and negative current for discharging;
- 314 Ah nominal capacity; and
- amp-hour retention in RTC backup registers while backup power is present.

The configured hard charge and discharge current limits are both 63 A. A full charge has successfully triggered the automatic 100% SoC calibration, and the installed pack has completed top balancing without an observed issue.

Static firmware settings contain the safety limits, monitor hardware template, and timing parameters. Companion-configurable values such as the shunt resistance, capacity, current polarity, balancing gate, diagnostics gate, and monitor topology are kept in the versioned STM32 runtime configuration in flash; they are not duplicated in the static settings.

The firmware separates active fault conditions and acknowledgement-required latches. The HV supervisor receives permission only while the BMS is running without blocking faults and holds one fresh, complete measurement set from the monitor chain.

Runtime faults keep measurement and monitoring active while preventing relay closure. An acknowledgement is accepted only after the underlying runtime condition has disappeared.

The STM32 also hosts:

- the dedicated HV supervisor;
- board-level analog and digital I/O;
- CAN communication;
- USB serial and the trusted text command/debug console; and
- the framed BMS v1 service interface over isolated UART and USB CDC.

3.1.1. Todo

High priority:

- Complete the as-built CubeMX configuration review; the pinout is considered correct.
- Calibrate and validate both isolated AMC3330 high-voltage measurements and their diagnostic inputs against external instruments.

- Exercise precharge, contactor pull-in/economizing, voltage-loss detection, opening, and recovery on the installed inverter. Characterize the persistent inverter DC-link voltage before assigning any welded-contactor interpretation.
- Complete and validate USB VBUS-sense behavior.
- Validate low-temperature charge inhibition and the remaining temperature fault/derating behavior on hardware.
- Continue long-duration GoodWe CAN observation and controlled charge/discharge validation, including communication-loss and recovery cases.
- Verify the implemented user-LED status language on the assembled hardware against `Documentation/architecture/home-bess-firmware-and-maintenance.md`.

Medium priority:

- Aux relay control (potentially useful for a future heating system).

3.1.2. GoodWe inverter CAN integration

OPERATIONAL / MONITORING

This section retains the evidence trail that led to the implementation and now also records the installation result. The selected 250 kbit/s GoodWe high-voltage profile is operational with the GW12K-ET-20 configured for the `Default *6` battery model in the Default 100 A group. The inverter recognizes the battery, reports the BMS measurements and limits in SEMS, and has completed real charge and discharge operation. The BMS remains the authority for the permitted current envelope. The CAN bus contains the STM32 and inverter; the ESP32 Gateway remains listen-only.

> **Important distinction**
>
> Two different families are being mixed in community material. A GoodWe high-voltage candidate uses 250 kbit/s and identifiers in the `0x453`– `0x460` range. A separate SMA/Pylontech-style profile uses 500 kbit/s and identifiers such as `0x351`, `0x355`, `0x356`, and `0x359`. The latter is not evidence that the former is used by the GW12K-ET-20.

3.1.2.1. Evidence status

Source	Relevant finding	Assessment
GoodWe GW6000-EH HV1 document	Reverse-engineered, non-official document. It specifies 11-bit CAN at 250 kbit/s, one-second-class updates, `0x453`–`0x458`, and additional `0x45A`/`0x460` frames. It also describes the inverter timeout frame `0x420` containing `GWBATTHV`.	Strongest public HV lead, but different model and not official.
ai-republic discussion 19	A GW5000-ES-20 user reported that newer G2 firmware rejected DIY BMS emulators. The author said a GoodWe HV specification worked for the GW6000-EH, but initially had no binding. The discussion does not prove an ET-20 frame map.	Important firmware/profile warning.
Current ai-republic HV descriptor and processor	The current code selects 250 kbit/s and sends `0x453`, `0x455`, `0x456`, `0x457`, and `0x458`. It does not parse an inbound request and does not send the `0x45A` or `0x460` frames listed by the reverse-engineered document.	Useful implementation lead; not target validation.
syssi discussion 114	The working trace uses the SMA/Pylontech-style family: 500 kbit/s, `0x359`, `0x351`, `0x355`, `0x356`, and `0x354`, with an inverter response at `0x305`. A later report says stale charge-disable state made the GoodWe unhappy.	Useful LV/SMA evidence, not HV ET evidence.

No official GoodWe public document found so far provides the complete BMS CAN frame map for the GW12K-ET-20. The official battery compatibility information establishes product and battery compatibility, but does not settle the bitrate, identifiers, payload byte order, or firmware-specific handshake.

3.1.2.2. Selected profile: GoodWe HV protocol

This is the implemented and hardware-validated profile for the GW12K-ET-20. Its byte-level interpretation still contains fields whose semantics are based on reverse engineering rather than an official GoodWe specification; those remaining uncertainties are tracked below and do not supersede the empirical compatibility result.

ID	Direction	Candidate payload	Confidence
`0x420`	Inverter -> STM32	Timeout/no-BMS beacon. ASCII `GWBATTHV` is `47 57 42 41 54 54 48 56`. Approximately one-second-class timing is reported; one copy states 1.3 s.	Medium for HV family; unconfirmed for ET-20.
`0x425`	Inverter -> STM32	Inverter-measured battery voltage and current. The public document says negative current means charging; the remaining bytes are not decoded.	Low/medium.
`0x453`	STM32 -> inverter	Number of 48 V battery modules. The document proposes integer nominal pack voltage divided by 48; that would produce a candidate value of 6 for this approximately 307 V pack.	Medium.
`0x455`	STM32 -> inverter	Two little-endian 16-bit bitmaps: BMS alarms followed by BMS warnings. Exact bit semantics remain provisional.	Medium.
`0x456`	STM32 -> inverter	Four little-endian 16-bit values: charge-voltage limit, charge-current limit, maximum discharge current, and minimum discharge voltage. Voltage/current resolution is 0.1 V/0.1 A.	Medium/high.
`0x457`	STM32 -> inverter	SoC and SoH as little-endian 16-bit values with 0.01% resolution, followed by four reserved bytes.	Medium/high.
`0x458`	STM32 -> inverter	Actual pack voltage, battery current, and average battery temperature as little-endian 16-bit values with 0.1 V, 0.1 A, and 0.1 °C resolution, followed by two reserved bytes.	Medium/high.
`0x45A`	STM32 -> inverter	Eight zero bytes in the reverse-engineered table. The document associates this frame with an inverter response at `0x425`.	Low/medium; validate before use.
`0x460`	STM32 -> inverter	Request flags related to enabling battery charging/discharging. The public document does not decode the flags.	Low; validate before use.

For this candidate, the actual-voltage field in `0x458` should use the measured pack voltage selected for FlexBMS, not a voltage inferred from SoC. The charge and discharge limits in `0x456` should be generated from fresh BMS limits. A loss of valid BMS data must never result in a larger permitted limit; the exact conservative fallback and whether a zero limit is accepted by this GoodWe firmware must be established during testing.

The current ai-republic sender is useful because it confirms the field layout above in executable code, but it is not a complete specification. In particular, it omits `0x45A` and `0x460`, does not parse `0x425`, and its warning-building code should be reviewed before reuse.

3.1.2.3. Candidate B: SMA/Pylontech-style CAN protocol, fallback only

This is the protocol described by the ESPHome/JK-BMS community work. It is a credible alternative only if the ET-20 firmware is configured for that battery profile. It should not be called the GoodWe HV protocol without evidence.

ID	Direction	Candidate payload
`0x351`	STM32 -> inverter	Charge-voltage limit, charge-current limit, discharge-current limit, and discharge-voltage limit; the published SMA-style mapping uses 0.1 V/0.1 A values.
`0x355`	STM32 -> inverter	SoC, SoH, and optional high-resolution SoC.
`0x356`	STM32 -> inverter	Actual battery voltage, current, and average temperature. The current is signed in the usual mapping, with positive commonly representing discharge.
`0x359`	STM32 -> inverter	Alarm and warning/status information. Exact bit interpretation must follow the selected SMA/Pylontech profile.
`0x354`	STM32 -> inverter	Present in the community trace, but its exact payload is not part of the current FlexBMS candidate contract.

0x305	Inverter -> STM32	Observed by the community implementation as an inverter response/ACK. This must not be assumed to be the HV handshake.

The evidence for Candidate B points to 500 kbit/s and a different byte-level contract from Candidate A. It is retained as a fallback because GoodWe G2 firmware and battery profiles may change the accepted protocol. It is not the first implementation target for the high-voltage ET-20 inverter.

3.1.2.4. Implemented software and installation result

IMPLEMENTED / HARDWARE-VALIDATED

The STM32 now selects the protocol through `Software/Master/Inc/Peripherals/GoodweCanConfig.h`. Candidate A is the default and selects 250 kbit/s; Candidate B selects 500 kbit/s. The CubeMX default and generated FDCAN timing use Candidate A, while the startup code applies the selected timing before starting the shared CAN driver.

The transmitter publishes the core frames once per second. Candidate A sends `0x453`, `0x455`, `0x456`, `0x457`, `0x458`, the eight-zero-byte `0x45A` diagnostic request, and the two-byte `0x460` default- battery compatibility frame with payload `00 00`. The latter was enabled for the inverter's `Default *6` profile after the core frames were accepted but the inverter continued to report `BMS: Discharge disabled`. `0x45A` is enabled to request the inverter's `0x425` measured voltage/current response during the diagnostic test; that response remains observation-only. Candidate B sends `0x351`, `0x355`, `0x356`, and `0x359`. Candidate B `0x354` remains disabled because its use is not sufficiently confirmed.

The CAN receiver passively records standard frames `0x420`, `0x425`, and `0x305`, including counters, timestamps, and the last payload. These frames do not receive application-level replies. The CAN peripheral handles the normal link-layer acknowledgement automatically. The reverse-engineered Candidate A material describes `0x420` as a timeout/no-BMS indication, not as a confirmed request requiring a response. The Candidate B material describes `0x305` as an inverter-originated periodic frame; community testing also treated it as an ACK or keepalive rather than a BMS request.

The STM32 publishes a 500 ms GoodWe diagnostic snapshot to Companion through the normal framed UART/Gateway path. Companion's Diagnostics page shows every Candidate A transmit frame's FDCAN-queue acceptance count and last-accept time, skipped cycles caused by an unavailable coherent BMS snapshot, the total and most recent transmit failure, the exact voltage/current last sent in `0x458`, and the raw `0x420`, `0x425`, and `0x305` receive counters, timestamps, and payloads. It also shows the hardware transmit/receive error counters, protocol state, bus-off/warning/error-passive flags, HAL error code, and TX FIFO free level so queue acceptance is not mistaken for confirmation of a completed CAN transmission. Browser-local CSV logging includes the same data, plus a clearly labelled candidate 0.1-unit decode of `0x425`; this decode is diagnostic evidence and not an input to battery safety or inverter limits.

On the installed inverter, adding the two-byte `0x460 00 00` compatibility frame removed the inverter's `BMS: Discharge disabled` condition for the selected `Default *6` profile. Immediately after the first firmware update, discharge followed the requested setpoint but intermittently returned to zero; the interruptions became more frequent before the inverter was returned to battery standby. After the subsequent restart and diagnostic firmware build, discharge remained stable during the available observation period. This is a useful compatibility result, but it does not yet prove whether `0x460`, `0x45A`, inverter state, or restart sequencing caused the transition. The diagnostics therefore remain enabled in normal builds because their runtime and bandwidth overhead is small and the installation has no separate USB/debug observation path during normal use.

The inverter responds promptly to EMS discharge setpoint changes. The initial DC-side observations were:

Requested	Observed `P_BAT`	Difference
0.5 kW	0.443 kW	−0.057 kW
2.0 kW	1.932 kW	−0.068 kW
4.0 kW	3.900 kW	−0.100 kW
6.0 kW	5.878 kW	−0.122 kW
10.0 kW	9.844 kW	−0.156 kW
12.0 kW	11.850 kW	−0.150 kW

These samples show a small offset and power-dependent difference. They are not yet a calibrated conversion between the GoodWe setpoint, DC battery power, AC inverter power, and net grid power. Charge response still requires an equally controlled test at a sufficiently low SoC.

Application-level responses are guarded by `GOODWE_CAN_ENABLE_APPLICATION_RESPONSES`, which defaults to `0`. It must not be enabled until a capture from the actual inverter identifies a required response and its payload.

CAN data is published only from a coherent BMS snapshot. Stale or invalid electrical or thermal measurements stop transmission; an unavailable SoC does not. A fresh snapshot with a common fault keeps the status frames alive but forces both directional current limits to zero. The configured under-temperature threshold is charge-only: it sets the charge current limit to zero while leaving discharge available if all common safety conditions remain healthy. For the EVE MB31 LiFePO4 pack, the inverter-facing charge target is fixed at 3.500 V per cell and charge current derates linearly from the configured maximum at 3.400 V maximum-cell voltage to zero at 3.500 V. The discharge-voltage target is 2.900 V per cell and discharge current derates linearly from the configured maximum at 2.900 V minimum-cell voltage to zero at 2.800 V. Lower current limits take effect immediately; increases recover at 10% of the configured maximum per second. These are soft, directional inverter limits and do not replace or latch the independent 3.600 V overvoltage and 2.500 V undervoltage BMS trips. Pack voltage uses the BCC cell-voltage sum. The Candidate A module count defaults to six and SoH defaults to 100% until a real SoH source exists. If no retained SoC calibration is available, the transmitter still publishes the coherent measurements and advertises 30% SoC until the automatic full-charge calibration completes; after that it publishes the calibrated SoC.

3.1.2.5. Open protocol questions

- Is `0x420` only a timeout indication on this firmware, and what are the exact timeout and recovery conditions?
- Is `0x460 00 00` independently required for discharge permission, or is the observed result dependent on `0x45A`, inverter state, or restart sequencing?
- Which remaining bytes in `0x425` are meaningful, and can its candidate voltage/current decode be fully validated across charge, discharge, and standby without using it as a safety input?
- What are the complete alarm/warning bit semantics in `0x455`, and which status transitions does the inverter retain internally?
- Does the inverter require the current one-second frame timing and burst ordering, or would other valid timing also remain stable?

3.1.2.6. Validation boundary

The installed software is a protocol-specific STM32 transmitter with passive receive diagnostics. The STM32 remains the safety authority: inverter-facing CAN frames communicate measured state and permitted limits, but they do not replace BMS fault evaluation, contactor permission, or HV supervision. Home Assistant may request an operating point only through the supported inverter control path; it must not write arbitrary CAN payloads.

The next evidence is long-duration operation plus controlled boundary tests: charge and discharge permission removal, BMS derating, CAN interruption, inverter timeout, restart, and recovery. Candidate B remains documented as a fallback, but it is not used by the installed system.

3.2. HV supervisor

CURRENT

The HV supervisor follows `OFF -> SELF_TEST -> PRECHARGE -> CONTACTOR_CLOSE -> RUN`. The transport-independent `setRunRequest(bool)` operation is implemented through the framed BMS v1 named service. It provides intent only; it does not command a contactor. The STM32 accepts operation only when its BMS and HV-supervisor conditions are healthy.

`SELF_TEST` checks BMS permission, measurement freshness, isolated-voltage diagnostics, agreement between the battery-side and cell-monitor voltages, an unenergized load side, and the available power source. `PRECHARGE` raises the load-side voltage and requires a stable voltage relationship before contactor pull-in. `CONTACTOR_CLOSE` transfers from the precharge path to contactor hold drive. `RUN` keeps monitoring the BMS permission and both HV measurements.

A removed request, lost BMS permission, or implausible HV feedback disables both drivers. A blocking fault does not erase a true run request: after an explicit user acknowledgement and successful STM32 revalidation, the BMS may resume the startup sequence. The request defaults off after an STM32 reset; loss of a Gateway or USB Companion transport does not alter it. The STM32 reports requested state, actual HV state, and any blocking reason separately.

USB-only operation supports service access but cannot energize the HV path. A recorded run request still requires healthy BCC and HV validation before the STM32 can start. The fitted hardware pull-down defines the inactive USB-sense level; the STM32 does not enable an internal pull.

> **Source of truth**
>
> Thresholds, timings, fault definitions, CAN identifiers, and register-level behavior are intentionally not duplicated here. The STM32 source code and hardware schematics are authoritative.

3.3. ESP32

CURRENT

The ESP32 is the network and telemetry controller in the isolated inverter-side domain. Its intended role is to observe BMS and inverter information, publish selected data to the local network, and support integration with the EMS and Home Assistant.

The ESP32 is not part of the battery safety chain. It is not allowed to transmit commands on the BMS CAN bus and must remain listen-only there. Loss of the ESP32 or Wi-Fi must not prevent the STM32 from protecting or isolating the battery.

`Software/Gateway` implements isolated UART v1, Wi-Fi provisioning/station/recovery, and the local compiled Companion maintenance page. The Companion provides stale-status diagnostics, fresh BMS monitoring, CSV logging, register reads, Gateway NTP RTC synchronisation and device-time readback, fault clear, immediate run-request actions, and station-LAN firmware updates. MQTT Home Assistant discovery is a current Gateway function. GoodWe CAN observations are produced by the STM32 and transported through the framed UART diagnostics; the Gateway itself remains listen-only on the shared BMS/GoodWe CAN bus. It is not a safety authority. The detailed transport, local-network, and recovery design is in `Documentation/architecture/home-bess-firmware-and-maintenance.md`.

After a station reconnect, the Gateway withdraws and re-announces `flexbms.local`; a temporary mDNS failure is retried. Gateway OTA boots the inactive slot as pending and confirms it only after the local HTTP/WebSocket service starts and a station address remains stable for ten seconds. Saved credentials are attempted directly first; bounded station retries continue even while the recovery AP is active. If stable connectivity cannot be established within three minutes, the ESP-IDF bootloader returns to the previous slot. The attempted version, rollback reason, and last Wi-Fi disconnect reason are retained for Companion diagnostics. Companion compares a content-derived web bundle identity and reloads its non-cacheable HTML shell automatically when a new bundle responds.

The local web service is intentionally bounded: it retains TCP capacity for MQTT and allows one queued telemetry frame per WebSocket. A browser that stops reading is retired after its first failed asynchronous send; a delayed callback from an old Companion socket cannot disturb its replacement connection.

The 4 MiB ESP32-C3 partition table uses two 1.5 MiB OTA application slots, a 512 KiB `bms_update` staging partition, and 384 KiB LittleFS reserved for future diagnostics/metadata. The committed Companion bundle is currently compiled into each application image, not served from LittleFS.

The Gateway application is written at the explicit `ota_0` offset in `Software/Gateway/partitions.csv` (currently `0x20000`), not ESP-IDF's generic `0x10000` default. Gateway release and normal PlatformIO upload tooling derive that address from the partition table; release creation verifies an ESP image magic byte at the derived address in the merged factory image.

3.3.1. Framed BMS protocol v1 contract

AGREED

The canonical byte-level implementation contract is `Documentation/protocol/uart-v1.md`. The following overview is retained in the system document; resolve any discrepancy in favour of the Markdown contract.

The STM32G491 BMS and ESP32-C3 Gateway communicate over isolated USART1 at 1 Mbit/s, 8-N-1, full duplex, without hardware flow control. Direct USB CDC uses the same framed bytes for the Web Serial Companion while retaining the trusted text command/debug console through a byte dispatcher. The STM32 is the safety authority. UART loss is a communication-health condition only: it must not force the HV path off or change the current run request. The `*!` BMS Companion protocol is retired.

Each endpoint sends a zero-payload heartbeat every 500 ms. A peer is declared lost only after 1.5 s without a complete CRC-valid frame. Invalid bytes or frames with a bad CRC do not refresh the timer. The Gateway and direct USB Companion share one STM32 service slot; a second request receives `BUSY` rather than being queued.

All multi-byte values are little-endian. Encoders and decoders write and read individual fields; they must not transmit native C or C++ structs. The maximum payload length is 512 bytes.

3.3.1.1. Frame

```
offset  size  field
0       2     MAGIC = 0x46 0x42 ("FB")
2       1     VERSION = 0x01
3       1     TYPE
4       1     SEQUENCE
5       2     LENGTH, uint16 payload length
7       n     PAYLOAD
7+n     4     CRC32, uint32
```

CRC is CRC-32/ISO-HDLC: polynomial `0x04C11DB7` (reflected `0xEDB88320`), reflected input and output, initial value and final xor `0xFFFFFFFF`. It covers the magic, header, and payload, but not the CRC field; the transmitted CRC word is little-endian. The standard check value is `CRC32("123456789") = 0xCBF43926`. On an invalid version, length, or CRC, the receiver discards the first magic byte and resumes magic scanning. The protocol uses no escaping or byte stuffing.

`SEQUENCE = 0` is reserved for heartbeat, telemetry, and events. A Gateway or USB Companion service request uses a value from 1 through 255; the STM32 response echoes that sequence.

3.3.1.2. Types

ID	Type	Direction
`0x01`	HEARTBEAT	both
`0x02`	STATUS	STM32 to Gateway or USB Companion
`0x03`	PACK	STM32 to Gateway or USB Companion
`0x04`	CELL	STM32 to Gateway or USB Companion
`0x05`	TEMPERATURE	STM32 to Gateway or USB Companion
`0x06`	HV_VOLTAGES	STM32 to Gateway or USB Companion
`0x10`	SERVICE_REQUEST	Gateway or USB Companion to STM32
`0x11`	SERVICE_RESPONSE	STM32 to Gateway or USB Companion
`0x12`	EVENT	STM32 to Gateway or USB Companion

3.3.1.3. Telemetry payloads

`HEARTBEAT` has an empty payload.

`STATUS` is 33 bytes:

```
bms_state:u8 | hv_state:u8 | flags:u16 | slave_count:u8 |
bms_active_errors:u32 | bms_latched_errors:u32 |
hv_active_errors:u32 | hv_latched_errors:u32 |
warnings:u32 | uptime_ms:u32 | soc_last_calibration_unix_s:u32
```

Status flag bits are: 0 BMS HV-ready, 1 charging allowed, 2 run request asserted, 3 complete measurements fresh, 4 isolated-UART Gateway peer alive, and 5 automatic balancing enabled. USB Companion heartbeats do not affect bit 4. Bit 6 is SoC valid, bit 7 current sensing enabled, and bit 8 the SoC calibration UTC value valid; bits 9–15 are zero. `slave_count` comes from the configured BMS monitor chain and is variable between builds. A production module contains eight monitor slaves, but a single-slave development chain is supported during bench work. The Gateway must use the reported value and must not reject an odd count.

`uptime_ms` is the STM32 HAL millisecond tick since its most recent reset. It is diagnostic telemetry only, resets to zero after a controller restart, and wraps naturally after about 49.7 days. The STM32 sends STATUS at least every 500 ms while its UART service runs.

`PACK` is 24 bytes:

```
pack_voltage_uV:u32 | pack_current_raw:i16 | soc_raw:u16 |
min_cell_uV:u32 | max_cell_uV:u32 |
min_ntc_raw:u16 | max_ntc_raw:u16 | min_ic_raw:u16 | max_ic_raw:u16
```

Voltage values are microvolts. Current is signed amperes times 64; positive current is charging. SoC uses the existing BCC raw range (`0 = -100%`, `65535 = 200%`), so percent is `100 * (soc_raw / 65535 * 3 - 1)`. The STM32 bounds its retained estimate and reported SoC to 0–100%, so the wider transport representation is not used in normal operation. NTC raw values convert as `raw / 65535 * 120 - 20` degrees C. IC raw values are centikelvin, so degrees C are `raw / 100 - 273.15`.

Use `pack_current_raw` only when STATUS bit 7 is set and `soc_raw` only when bit 6 is set. SoC is informational and never commands HV. The estimate is retained in the CR2032-powered RTC backup domain with a version marker and checksum; an erased or corrupt value is invalid. Current sensing is deliberately disabled in the single-slave development configuration, so current and SoC are unavailable there. Production enables it only on CID 1. Full-charge calibration requires fresh measurements with no active BMS error, every cell at or above 3.450 V, current from −0.100 A through C/50 (6.28 A for the 314 Ah default), continuously for 300 s. Any failed condition restarts the timer; calibration sets SoC to 100%. The 3.500 V-per-cell inverter charge target deliberately remains above the 3.450 V minimum-cell qualification threshold, so calibration stays reachable while maximum-cell current derating and balancing prevent normal charging from reaching the 3.600 V hard trip. Coulomb counting is bounded at 0% and 100%, allowing a subsequent discharge to lower SoC normally without retaining an out-of-range value. When valid UTC is available, the STM32 persists that instant in integrity-protected backup registers and reports it in `soc_last_calibration_unix_s`; it otherwise reports no calibration time.

`HV_VOLTAGES` is 12 bytes:

```
valid_flags:u8 | reserved:u8[3] | bat_plus_uV:u32 | load_plus_uV:u32
```

Bit 0 of `valid_flags` means both values came from one fresh, coherent STM32 ADC/DMA scan with a valid ADC reference. Bits 1–7 and the three reserved bytes are zero. The values are positive microvolts from the isolated AMC3330 battery-side (BAT+) and load-side (LOAD+) channels. When the valid bit is clear, consumers show the values as unavailable rather than treating the zero payload as a measurement. These diagnostic readings are sent with every normal fresh snapshot and do not change PCC thresholds, fault decisions, or STM32 safety ownership.

`CELL` is 51 bytes, once per configured slave:

```
slave_index:u8 | balance_mask:u16 | cell_voltage_uV[12]:u32
```

`slave_index` is zero-based. `module_index = slave_index >> 1` and `slave_in_module = slave_index & 1`. Balance-mask bit 0 represents cell 0 through bit 11 for cell 11; bits 12–15 are zero. Each slave always has twelve cells. In a single-slave development chain, only index 0 is sent; it maps to module 0, slave 0, and no partner frame is implied. `balance_mask` reports the cells selected for the current balancing pulse after the BCC driver-status check succeeds. It remains stable across brief measurement pauses and therefore does not represent the instantaneous state of the bleed switches.

`TEMPERATURE` is 11 bytes, once per configured slave:

```
slave_index:u8 | ntc_raw[4]:u16 | ic_temp_raw:u16
```

The four NTC values and one IC value are included together, making this packet fixed-size. This matches the installed four-NTC-per-slave configuration and must be enforced as a compile-time configuration check in both endpoint implementations.

The STM32 sends STATUS and a complete HV_VOLTAGES/PACK/CELL/TEMPERATURE snapshot when fresh measurement data is available, no more often than once every 500 ms. It sends EVENT immediately on a change. The Gateway must treat HV_VOLTAGES, PACK, CELL, and TEMPERATURE data as invalid while STATUS says measurements are not fresh.

3.3.1.4. States, faults, and events

BMS states are `0 STARTING`, `1 READY`, `2 RUNNING`, `3 ERROR`, and `4 CRITICAL`. HV states are `0 OFF`, `1 SELF_TEST`, `2 PRECHARGE`, `3 CONTACTOR_CLOSE`, and `4 RUN`. `CRITICAL` has no bitmap: it represents an unrecoverable condition in the current software and forces HV safe-off for the remainder of that boot.

BMS ERROR bitmap bits are: 0 `CONFIGURATION_INVALID`, 1 `SLAVE_UNAVAILABLE`, 2 `BCC_DIAGNOSTICS`, 3 `CELL_VOLTAGE_LIMIT`, 4 `THERMAL_LIMIT`, 5 `CURRENT_LIMIT`, 6 `BCC_INTEGRITY`, 7 `ADC_FAULT`, 8 `BALANCING_HARDWARE_FAULT`, and 9 `BCC_COMMUNICATION`. Bits 10–31 are reserved and zero.

HV ERROR bitmap bits are: 0 `HV_SENSOR_DIAGNOSTIC`, 1 `BATTERY_VOLTAGE_MISMATCH`, 2 `LOAD_SIDE_ENERGISED`, 3 `PRECHARGE_TIMEOUT`, 4 `PRECHARGE_VOLTAGE_LOST`, and 5 `CONTACTOR_VOLTAGE_LOST`. Bits 6–31 are reserved and zero.

warnings is non-latched. Bits are: 0 WATCHDOG_RESET, 1 STARTUP_DIAGNOSTICS_BYPASSED, and 2 BATTERY_VOLTAGE_MISMATCH_OFF. Bits 3–31 are reserved and zero. WATCHDOG_RESET is a recorded warning that an accepted acknowledgement clears. STARTUP_DIAGNOSTICS_BYPASSED remains present while startup diagnostics are disabled in runtime configuration; BATTERY_VOLTAGE_MISMATCH_OFF remains present while the measurements disagree. WATCHDOG_RESET is set when this STM32 boot followed an independent or window watchdog reset. Gateway liveness, CAN condition, near-limit indication, and disabled automatic balancing are not BMS warnings in this release.

The STM32 compares BAT+ and the slave-reported pack voltage continuously. A disagreement is an OFF-state warning while no run request is present; a run request promotes the same condition to the blocking BATTERY_VOLTAGE_MISMATCH HV error. Releasing the request deasserts that active HV error, retains its latch for acknowledgement, and restores the warning until the readings agree.

An active ERROR is present now. A latched ERROR remains blocking after its live condition clears until the STM32 accepts acknowledgement. The STM32 Fault Manager is the sole owner of active/latched aggregation, acknowledgement, and immediate HV safe-off. Detection modules may only assert or deassert their assigned condition. Startup monitor-chain, register-initialisation, and diagnostics attempts use their source retry budget before asserting their ERROR input. Once exhausted, they use these ordinary ERROR semantics.

A live TPL/BCC communication loss immediately safe-offs HV and re-enters the same non-blocking TPL, CID, register, and measurement initialisation path. A successful complete measurement and fault-status cycle deasserts the active communication error; its latched error still requires acknowledgement.

The independent watchdog starts after the required clock and DMA setup, before every fallible peripheral initialisation, has a nominal 500 ms timeout, and is refreshed every 100 ms only while both the PCC loop and BCC task continue to make progress. BCC initialisation retry waits are non-blocking, so they do not mask a stalled task. A subsequent platform-initialisation failure directly disables the precharge output and contactor PWM before the watchdog reset; the contactor driver pull-down is the hardware fallback. A clock-initialisation failure may occur before the watchdog is available; the hardware pull-down still holds the contactor control line safe.

The STM32 ADC is calibrated at startup. TIM7 triggers a three-rank HV scan at 50 Hz: load-side differential voltage, battery-side differential voltage, and VREFINT. Each rank uses 640.5 ADC clock cycles of acquisition and 256x oversampling with right-shift 4. DMA publishes complete scans as one snapshot. The STM32 drives VREF+ from its 2.9 V internal reference buffer. ADC/DMA errors and a 2.8–3.0 V ADC reference are checked every 20 ms. DMA progress is stale only after no completed scan for 100 ms. Three consecutive failed checks assert ADC_FAULT and immediately safe-off HV. Calibration is not repeated during normal operation.

EVENT is five bytes:

```
event_id:u8 | value:u32
```

ID	Event	Value
0x01	BMS state changed	new BMS state
0x02	HV state changed	new HV state
0x03	BMS active errors changed	new active mask
0x04	BMS latched errors changed	new latched mask
0x05	HV active errors changed	new active mask
0x06	HV latched errors changed	new latched mask
0x07	warnings changed	new warning mask
0x08	measurement freshness changed	0 or 1

EVENT is a convenience notification; STATUS is authoritative, so loss of an event is harmless.

3.3.1.5. Services

The service-request payload is service_id:u8 | arguments...; the service-response payload is service_id:u8 | result:u8 | response_data.... Result values are 0 OK, 1 DENIED, 2 INVALID, 3 BUSY, and 4 USB_HOST_ACTIVE. INVALID covers an unknown service, bad request length, bad argument, or nonexistent slave index. DENIED means a valid request cannot safely be performed in the current STM32 state. BUSY means another named service is still in progress. USB_HOST_ACTIVE means an enumerated USB host prevents the STM32 ROM bootloader from accepting USART1. OK means that the STM32 accepted and invoked the requested operation; STATUS and EVENT show the resulting operating state.

ID	Service	Request arguments	OK response data
0x01	GET_STATUS	none	33-byte STATUS
0x02	SET_RUN_REQUEST	requested:u8 (0 or 1)	none
0x03	ACKNOWLEDGE_FAULTS	none	none
0x04	READ_REGISTER	slave_index:u8, register:u8	slave_index:u8, register:u8, value:u16
0x05	SET_RTC	unix_time_s:u32 UTC	none
0x06	GET_DEVICE_INFO	none	firmware_version:u32
0x07	PREPARE_STM32_BOOTLOADER	firmware_version:u32, image_length:u32, image_crc32:u32	none
0x08	GET_RTC	none	unix_time_s:u32 UTC
0x09	SET_BALANCING_ENABLED	enabled:u8 (0 or 1)	none
0x0A	COMMIT_STM32_BOOTLOADER	none	none (successful request enters ROM immediately)
0x0B	GET_CONFIG	none	18-byte runtime configuration status and values
0x0C	SET_CONFIG	slave_count:u8, current_sense_slave:u8, shunt_resistance_uohm:u32, battery_capacity_mah:u32, invert_current:u8, balance_enabled:u8, startup_diagnostics:u8	none; the STM32 resets after responding OK
0x0D	GET_DIAGNOSTIC_REPORT	slave_index:u8	retained startup diagnostic report for that BCC

SET_RUN_REQUEST(0) immediately removes the request through the STM32 PCC path. A request of 1 does not guarantee an HV start: the STM32 alone decides whether BMS and HV checks permit the sequence. ACKNOWLEDGE_FAULTS is denied while any ERROR condition remains active or the BMS is CRITICAL. It is also denied when no inactive BMS/HV ERROR latch or recorded WATCHDOG_RESET warning remains to clear. Accepted acknowledgement clears those states only, leaving live and configuration warnings visible until their underlying condition changes; it deliberately preserves the run request. PCC can therefore restart from OFF immediately once the Fault Manager permits it. Neither the Gateway nor Home Assistant can bypass a fault, write BCC registers, or override the HV supervisor. firmware_version is packed as major | (minor << 8) | (patch << 16) | (build << 24).

SET_BALANCING_ENABLED controls the STM32-owned automatic balancing enable gate and persists the value in runtime configuration. It applies immediately after the flash write succeeds and does not reboot the STM32. An enabled value does not guarantee that any cell is balancing: the STM32 still requires a running, fault-free BMS, fresh measurements, and its configured cell-voltage and current thresholds. For the MB31 pack, top balancing starts at 3.400 V and a 10 mV difference from the pack-wide minimum cell. A selected cell continues down to a 5 mV difference, providing hysteresis. Balancing is permitted only from −0.100 A through C/10 (31.4 A at 314 Ah), so meaningful discharge inhibits it. Thirty-second pulses are used and all 12 connected cells per slave may be selected at once. The highest-voltage eligible cells receive priority; rotation is used only to share equal-voltage ties. With all channels active, one module board dissipates approximately 6.2 W in its bleed resistors. Before each pulse, the STM32 disables the global drivers, clears every retained CB channel, programs the complete desired mask, and enables the global drivers last. It then verifies the actual driver-status mask. Leaving the running/eligible state disables balancing immediately. A disabled value inhibits the BCC balancing drivers on the next BCC loop. New configuration defaults enable balancing. The configuration schema version is 3. Version-2 records are accepted with startup diagnostics enabled as the safe legacy default and are upgraded to version 3 when next saved through SET_CONFIG. startup_diagnostics defaults to 1; setting it to 0 skips only the startup BCC diagnostics routine for debugging or bench testing, while all other measurement, fault, balancing, and HV safety checks remain active.

GET_DIAGNOSTIC_REPORT returns the slave index, BCC CID, failed-check bitmask, low-level status code, and the diagnostic that was executing when a driver failure occurred. Register context, the attempt number and raw/checked GPIO masks are included in extended reports. The first failure per slave is held in STM32 RAM until restart; otherwise the latest successful report is available. Reports are historical, while Status identifies live faults. GPIO open-terminal diagnostics only fail for configured temperature inputs; unused GPIOs remain visible in the raw mask. Failed startup

diagnostics repeat CID and register initialization after the retry delay so communication resets cannot invalidate the next attempt.

`PREPARE_STM32_BOOTLOADER` and `COMMIT_STM32_BOOTLOADER` form the STM32-update handoff. The Gateway must already have staged and CRC-checked the image. The STM32 validates the prepare request shape and image length; `firmware_version` and `image_crc32` identify the Gateway-staged image but cannot be verified by an STM32 that does not receive its bytes. It returns `DENIED` unless the run request is off and it can de-energise the HV path. It returns `USB_HOST_ACTIVE` when its USB CDC device is enumerated. Prepare immediately holds HV safely off and blocks normal services, then returns `OK`. The prepared state expires after 3 seconds unless the Gateway sends commit. Commit has no response: after successfully transmitting it, the Gateway owns USART1 and immediately changes to ROM settings. If prepare or commit transmission fails, the STM32 remains in (or returns to) normal BMS operation; the Gateway reports failure rather than waiting indefinitely. It then performs ROM bootloader sync, transfer, readback verification, and `Go` directly on USART1; these are not FlexBMS UART messages. On return, the Gateway waits for an application heartbeat before reporting completion. Any failure after erase has begun requires wired STM32 recovery; the Gateway rejects further STM32 OTA attempts until it observes an application heartbeat. There are deliberately no update-data, retry, or status frame types.

3.3.1.6. Test vectors

The following complete frames are hexadecimal, including the little-endian CRC:

```
HEARTBEAT
46 42 01 01 00 00 00 8F 7A FB 7D

GET_STATUS, sequence 0x2A
46 42 01 10 2A 01 00 01 8F 21 B2 4B

SET_RUN_REQUEST(true), sequence 0x2B
46 42 01 10 2B 02 00 02 01 16 26 B1 DC

READ_REGISTER response, sequence 0x2C, slave 3, register 0x20, value 0x1234
46 42 01 11 2C 06 00 04 00 03 20 34 12 1A 45 F6 9C

PREPARE_STM32_BOOTLOADER, sequence 0x2D, version 1.2.3 build 4,
image length 131072 bytes, image CRC32 0xA1B2C3D4
46 42 01 10 2D 0D 00 07 01 02 03 04 00 00 02 00 D4 C3 B2 A1 42 C9 9E 95

COMMIT_STM32_BOOTLOADER, sequence 0x2E
46 42 01 10 2E 01 00 0A 50 6F 02 53
```

3.3.2. Pinout

The Gateway uses an ESP32-C3-WROOM-02U-N4. The module pin names and numbers below follow the Espressif datasheet; net names follow the Master schematic. GPIO17 is used internally for module flash and is not exposed.

Module pin	GPIO / module name	Net	Use
1	3V3	`p3v3_INV`	3.3 V supply
2	EN	`EN`	Chip enable
3	GPIO4 / IO4	`CAN_TX`	CAN controller transmit
4	GPIO5 / IO5	`CAN_RX`	CAN controller receive
5	GPIO6 / IO6	`RS485_RO`	RS-485 receiver output
6	GPIO7 / IO7	`RS485_DI`	RS-485 driver input
7	GPIO8 / IO8	`IO8/STRAP`	Strapping pin
8	GPIO9 / IO9	`IO9/BOOT`	Boot-mode strap
9; EPAD 19	GND	`GND_INV`	Ground and module ground pad
10	GPIO10 / IO10	`ESP_VBUS_SENSE`	USB VBUS sense
11	GPIO20 / RXD	–	Not connected; UART0 RX unused
12	GPIO21 / TXD	–	Not connected; UART0 TX unused

Module pin	GPIO / module name	Net	Use
13	GPIO18 / IO18	USB_ESP32_N	USB D-
14	GPIO19 / IO19	USB_ESP32_P	USB D+
15	GPIO3 / IO3	UART_ESP_RX	STM32 USART1 TX input
16	GPIO2 / IO2	UART_ESP_TX	STM32 USART1 RX output; strapping pin
17	GPIO1 / IO1	USR_LED	Gateway user/status LED
18	GPIO0 / IO0	RS485_EN	RS-485 driver enable

3.4. Energy management system

SHADOW OPTIMIZATION OPERATIONAL

EMHASS runs on Home Assistant OS as the supervisory optimizer. It combines battery state, site demand, PV production, inverter limits, dynamic electricity prices, and eventually the EV charger, heat pump, and air conditioners. The present deployment calculates and publishes schedules but does not actuate the inverter.

The detailed real-time controller specification is maintained in the separate Zaalsteden project, covering both Zaalsteden and Veldkamp. See Controller documentation reference for its project-relative document index. The implementation notes below record the earlier EMS baseline; the controller specification records the decisions agreed on 9 October 2026, including feeder topology, EV priority, telemetry validity, deployment, and the accepted communications-outage limitation. Controller software and automatic actuation remain unimplemented.

The EMS may request charging or discharging, but it must operate inside limits supplied by the BMS and inverter. It must tolerate missing telemetry and fall back to a zero-battery-power state. The STM32 and GoodWe remain responsible for their own safety and derating; an EMHASS result is never permission to exceed either device's current limit.

3.4.1. Installed energy-system topology

Element	Installed baseline
Battery	96s / 314 Ah LiFePO4; approximately 96 kWh configured in EMHASS. Desired operating range 10–90% SoC, with no reserved emergency-backup capacity. The BMS hard current rating is 63 A in both directions.
Hybrid inverter	GoodWe GW12K-ET-20 with battery and 30 x 370 Wp PV: 15 panels south-west and 15 north-east, both at approximately 15 degrees.
Second PV inverter	GoodWe GW15K-DT with 36 x 300 Wp south-east PV at approximately 35 degrees. A dedicated three-phase Modbus meter reports this inverter every five seconds.
Grid connection	Three phase, 3 x 25 A. The current aggregate model uses 17.25 kW in each direction; protecting every individual phase remains separate work.
Flexible demand	EV charging, heat pump, air conditioners, and other house loads are included in measured house demand but are not yet individually optimized. The battery, heat pump, and EV charger also share a provisional 3 x 20 A installation constraint whose exact scope still needs refinement.

The GW12K-ET-20 is reached through a GoodWe WiFiLAN Kit-20 at `192.168.1.160`; the observed dongle firmware is `V2.7.64`. SEMS also contains the separate GW15K-DT. The hybrid inverter has no GoodWe house/load meter, so SEMS load power must not be used as the site-load source.

3.4.2. Home Assistant measurements and controls

Entity or source	EMS use
`sensor.flexbms_state_of_charge`	Live initial battery SoC.

Entity or source	EMS use
`sensor.flexbms_pack_power`	DC-side BMS power feedback. The BMS convention must be converted explicitly when compared with EMHASS and GoodWe signs.
`sensor.schuur_gw12k_et20_active_power`	Hybrid-inverter AC power feedback.
`sensor.smart_meter_net_power`	Total site import/export at the P1 meter.
`sensor.p1_meter_power_phase_1` through `_3`	Per-phase grid power used for the future 3 x 25 A guard.
`sensor.rf24mesh_bijkeuken_zonnepanelen_bijkeuken_vermogen`	Accurate AC power from the dedicated meter after the 15 kW PV inverter.
`sensor.emhass_total_pv_power`	Home Assistant aggregate PV-power input for EMHASS.
`sensor.emhass_house_load_power`	Home Assistant site-load input, including the EV charger, heat pump, and air conditioners.
`sensor.emhass_import_price` `sensor.emhass_export_price`	Current price plus timestamped forecast attributes supplied to EMHASS.
`select.schuur_gw12k_et20_ems_mode`	GoodWe EMS mode: tested manual charge, discharge, and battery-standby selection.
`number.schuur_gw12k_et20_ems_power_limit`	GoodWe EMS power request in watts.

The inverter entities currently come from the custom GoodWe SEMS Home Assistant integration. Its measurements and control entities work, but the transport depends on the integration and SEMS behavior. Before closed-loop actuation, direct local GoodWe access through the WiFiLAN Kit-20 should be evaluated. Local control is preferred if it exposes the required writes reliably because it reduces delay and removes cloud availability from the control loop; the working SEMS path can remain a fallback and monitoring source.

3.4.3. Measured shared feeder loss

A measured voltage difference of approximately 7 V phase-to-neutral per phase was observed between 0 W and 12 kW balanced three-phase transfer through the long, relatively high-ohmic feeder between the junction box and the main electrical cabinet. The feeder is shared by the hybrid inverter, EV charger, heat pump, and other loads. This is an observed endpoint voltage drop, not yet a validated loss curve. The loss therefore depends on the aggregate net current through the feeder (approximately an I^2R effect), not on battery power alone.

At the 12 kW operating point the measured loss is approximately 350 W in total, which explains the observed increase of house-load power from approximately 350 W to 700 W. EMHASS currently uses unity AC/DC efficiency values in the MVP configuration, but this loss must not be assigned solely to inverter efficiency. It should be represented against the aggregate feeder current or net junction-box power. Future validation should record phase currents, endpoint voltages, EV/heat-pump demand, inverter AC power, and P1 power at several operating points and in both directions.

3.4.4. Current EMHASS MVP

SHADOW ONLY

The current configuration uses:

- EMHASS `naive-mpc-optim` with a 15-minute optimization step;
- a live `soc_init` and a 96 kWh nominal battery model;
- 10% minimum and 90% maximum SoC;
- 12 kW maximum battery charge and discharge power, with 95% efficiency in each direction;
- a profit objective with dynamic import and export price forecasts;
- a 17.25 kW aggregate grid import/export limit;
- a schedule run four times per hour, followed by publication of battery, grid, PV, load, SoC, price, and optimizer-status forecast entities; and
- no inverter actuation: the schedule is observed as a shadow plan only.

Battery charge/discharge wear weights, SoC dwell penalties, inverter stress cost, and capacity-charge cost are deliberately zero in the MVP. An arbitrary wear price would change dispatch without yet being tied to measured degradation or a chosen operating policy. Throughput and cycle behavior should first be measured; a defensible marginal wear cost can then be introduced if schedule churn or low-value cycling is observed.

The current house-load forecast is a temporary 96-quarter-hour bootstrap profile reconstructed from available measurements. This avoided an EMHASS shape error while the new aggregate load sensor did not yet contain a complete Recorder day. It must be replaced by a history-based weekday/time model after sufficient clean history exists. The three PV orientations are represented in the EMHASS model, but total-PV forecast accuracy has not yet been calibrated against both installed inverter measurements.

Nord Pool currently supplies the market-price horizon. Home Assistant template sensors add the applicable Zonneplan import/export tariff components before passing prices to EMHASS. The Zonneplan account/integration should be revisited when the new contract is active. The 2026 calculation assumes the installation remains an annual net importer under the current net-metering treatment. The price templates and optimization interpretation must be reviewed before the Dutch net-metering rules change on 1 January 2027, when taxes on imported energy are no longer offset in the same way by exported energy.

3.4.5. Horizon and terminal battery value

`soc_final` is a hard equality in the current EMHASS model. An early shadow run incorrectly set it to 90%, which forced a 12 kW recharge in the final expensive interval simply to meet the endpoint. The current MVP uses 50%. This permits economically useful discharge while retaining 48 kWh at the end of the known forecast, but 50% is still an arbitrary boundary condition rather than a physical reserve requirement.

The next practical strategy is a fixed 48-hour rolling horizon:

- use actual Nord Pool/Zonneplan prices for every published interval;
- extend the unknown tail with quarter-hour estimates derived from comparable historical weekdays, including the applicable tariff components;
- provide load and PV forecasts over the same complete horizon;
- retain the 50% hard target only at the distant end of that fixed horizon; and
- execute only the first interval before solving again with fresh data.

This prevents the horizon from shrinking to the last published price and moves the artificial endpoint away from the near-term dispatch decision. A later, mathematically cleaner improvement is a terminal salvage value: reward each kWh remaining at the endpoint according to its expected future replacement value, while leaving final SoC free inside 10–90%. EMHASS does not currently expose that objective directly. Implementing it would require an EMHASS enhancement or a separate optimizer; the hourly SoC deficit/surplus penalties are battery-dwell preferences and are not equivalent to a terminal value.

3.4.6. Path to automatic inverter control

Automatic control must not be enabled until the shadow output and feedback paths have passed the observation gates below. The executor should then:

1. read the current `P_batt_forecast` command and reject stale or unsuccessful plans;
2. convert the EMHASS sign convention explicitly into `Charge Battery` or `Discharge Battery` plus a non-negative GoodWe power limit;
3. clamp the request to the inverter's 12 kW limit; GoodWe must continue to enforce the fresh BMS charge/discharge limits received over CAN, and those limits should also be exposed to Home Assistant if the executor is to pre-clamp against them;
4. apply SoC bounds, a small zero-power deadband, minimum mode residence time, and rate limiting to prevent unnecessary mode chatter;
5. calculate remaining headroom on each P1 phase before allowing battery, heat-pump, or EV power to approach either the 3 x 25 A grid connection or the provisional shared 3 x 20 A installation limit;
6. verify response using BMS power, hybrid-inverter AC power, and P1 power; and
7. return to a tested zero-battery-power state on stale data, failed writes, Home Assistant/EMHASS unavailability, or manual override.

Battery standby is the functional zero-power state for the first executor. Idle-energy optimization is a separate unresolved topic. Battery standby has been observed to consume approximately 70–75 W, sometimes from the battery. ECO mode has also shown approximately 5 W in one sequence, but returning from an EMS charge/discharge

request to standby did not consistently restore that low-power condition. Selecting another operating mode and then ECO changed the result, and opening the BMS contactor did not produce the expected increase in grid power. This behavior is dynamic and must be characterized before ECO or contactor control is included in normal EMS sequencing.

3.4.7. Observation and improvement plan

Area	Record	Decision after evidence
Optimizer	Run status, horizon start/end, objective, schedule changes between runs, final-interval power, and SoC endpoint.	Detect terminal-boundary artifacts, infeasibility, timestamp/DST errors, and excessive schedule churn.
Forecasts	15-minute predicted versus measured house load and PV; bias and absolute error by time of day and weekday.	Replace the bootstrap load, tune PV models, and choose historical-tail rules.
Prices	Published interval count, timestamps, import/export values, tariff components, and update time.	Catch missing tomorrow prices, unit/tax mistakes, and the required 2027 tariff transition.
Power response	Requested GoodWe limit, EMS mode, `P_BAT`, inverter AC power, P1 response, phase currents, endpoint voltages, settling time, cable drop, and residual error.	Fit charge and discharge mappings, current-dependent cable/conversion losses, deadband, and closed-loop correction if necessary.
Battery model	SoC change versus integrated Wh, charge/discharge efficiency, usable capacity, temperature, and BMS derating.	Calibrate capacity/efficiency and enable EMHASS battery identification first in observation mode.
Grid protection	Maximum and duration of current/power on each P1 phase, including EV, heat-pump, 3 x 25 A service, and provisional shared 3 x 20 A loading.	Confirm the actual circuit scope, then set conservative per-phase headroom, transient allowance, and load-shedding priorities.
CAN	Transmit-cycle/skipped-snapshot counts, per-frame timestamps, TX failures, TEC/REC, bus state, and normalized `0x458` versus `0x425` current.	Investigate gaps, sign/scaling drift, permission loss, bus recovery, and any recurrence of intermittent zero power.
Modes and idle	Mode transition sequence, BMS contactor state, battery/grid power, and steady-state auxiliary consumption.	Define a repeatable zero-power/ECO sequence or keep battery standby as the reliable fallback.
Economics	Actual import/export cost, battery throughput, conversion loss, and a no-battery baseline.	Quantify savings and introduce a wear cost only if supported by the observed value and cycling.
Availability	Age and availability of BMS, P1, PV, GoodWe, price, optimizer, and command entities; HA/add-on restarts and failed writes.	Define timeouts, alarm thresholds, and the exact automatic fallback behavior.

After approximately one week of clean shadow data, review timestamp alignment, price completeness, load/PV error, endpoint behavior, and plan stability. After several representative charge/discharge cycles, estimate setpoint mapping, efficiency, usable capacity, and SoC drift. Repeat the review after several weeks and after materially different weather or heating demand; the large battery and flexible household loads make a single sunny test day insufficient. Automatic actuation should start with conservative power and a supervised trial, then expand only after the per-phase fuse guard and stale-data fallback have been exercised.

3.5. Home Assistant integration

CURRENT

Home Assistant is the intended user-facing integration point for the installation. It can provide:

- battery, inverter, PV, grid, and load dashboards;
- fault and availability notifications;
- manual operating-mode selection;
- automation around dynamic prices and flexible loads; and
- long-term energy and temperature history.

The current Home Assistant OS host is a Raspberry Pi 5. The BMS MQTT entities, GoodWe entities, Nord Pool data, aggregate PV/load templates, price templates, and EMHASS forecast entities are installed and exposed. The GoodWe integration does not provide site house/load power because neither inverter has the required GoodWe load meter; the P1 and dedicated PV measurements remain authoritative for site accounting.

Home Assistant must not become a safety dependency. If Home Assistant is unavailable, the STM32 must continue to protect the battery and the EMS must remain within the last valid limits or stop requesting power.

The Gateway publishes one retained MQTT Device Discovery record and retained compact BMS state below a MAC-derived `flexbms/flexbms_XXYYZZ` prefix. It exposes pack voltage/current/power, SoC when valid, cell and temperature extrema, BMS/HV state, active faults, freshness, UART health, and the STM32-owned run-request switch. Minimum and maximum cell voltage display in V with three decimals; cell delta displays as whole mV. It also exposes separate diagnostic uptime values for the ESP32 Gateway and STM32 BMS controller. `ON` and `OFF` requests share the existing single Gateway-to-STM32 service slot; the next STM32 status is authoritative and never promises contactor closure. Broker settings (host, port, username, write-only password) are configured only through the trusted station-LAN Companion page. The binary freshness, run-state, fault, and UART-health entities publish canonical MQTT `ON`/`OFF` states. Fault acknowledgement, balancing, firmware upload, registers, raw diagnostics, and inverter/EMS power control remain outside Home Assistant MQTT v1.

An optional Gateway-local RS485 extension reads zero, one, or two SDM72D-M-2 meters. Both slots are disabled by default. The Companion configures each meter's name, slave address, and reverse power direction, plus shared baud rate (default 9600), parity, and stop bits. UART0 uses GPIO7 TX, GPIO6 RX, and GPIO0 driver enable independently of the STM32 UART1 link. The Gateway reads phase currents, phase and total signed active power every second, and import/export kWh every 30 seconds. Each enabled slot gets a separate HA device with stable Gateway-and-slot identity. Failed or stale reads become unavailable rather than zero; non-retained electrical and energy samples expire independently. Reversing direction negates active power and exchanges import/export energy while preserving current magnitudes. The extension performs no inverter regulation; Home Assistant owns that control. Settings persist in Gateway NVS and are editable only on the trusted station LAN. Register offsets, wiring, freshness, and commissioning checks are documented in `architecture/rs485-energy-meters.md`.

3.6. Companion application

CURRENT

`Software/Companion` is the FlexBMS-owned BMS-only application with two web targets: a direct USB Web Serial target for independent service diagnostics and an ESP32-hosted WebSocket target for local maintenance. A portable, unsigned Windows executable packages the direct-USB target in a thin Electron/Chromium wrapper. It has no installer, updater, or native serial-protocol implementation.

Both targets provide stale-status explanation, live fresh telemetry, CSV logging, named service requests, device-time readback, and read-only register inspection. The Gateway obtains UTC from NTP and shows its latest successful RTC sync in the browser locale; direct USB shows that NTP status is Gateway-only. The Gateway target also shows Wi-Fi and UART state and, only in connected station mode, single-bundle firmware updates. The direct-USB target exposes neither a raw terminal nor firmware-update UI; Gateway requests use an explicit allowlist.

`scripts/build-release.ps1` interactively confirms the release version, updates the Companion package version when a new one is entered, and builds the Companion, Gateway, and STM32 artifacts with checksums and one `FlexBMS_bundle.fbu` OTA package containing both targets, versions, byte lengths, and CRC-32 values. `-Version` supports unattended use. Each release includes `flash-release.ps1`: ST-Link/SWD flashes the STM32 image, while a selected ESP32-C3 COM port flashes a complete Gateway factory image. Both actions start immediately after checksum verification. Repeating an existing version requires an explicit choice and creates a timestamped output without overwriting the prior release.

4. Operating concept

4.1. Start-up

At start-up, the STM32 validates its configuration and initializes the complete TPL chain. It assigns each battery monitor a chain address, configures its registers, runs diagnostics, and obtains a fresh measurement set. Both HV

outputs remain off. After BMS validation, the separate HV supervisor can run its self-test and precharge sequence only when a new run request is supplied.

4.2. Normal operation

During normal operation the STM32 repeatedly measures the pack and evaluates its safety state. Battery limits and status are made available to the surrounding system. The inverter converts energy between the DC battery and the AC installation. EMHASS currently calculates the desired operating point in shadow mode; after validation, a separate executor will translate the first schedule interval into a bounded GoodWe request.

4.3. Faulted operation

A battery, measurement, or communication fault immediately removes relay permission. Monitoring continues where possible so that the live condition can be observed. Faults remain latched until they are acknowledged through the supported service path and the BMS has successfully revalidated the battery chain.

The BMS exposes the HV supervisor as one aggregate blocking fault while the supervisor retains separate active, latched, and historical HV reasons. The Companion clear command can clear an HV latch only when the live HV conditions are healthy. The BMS then repeats chain initialization, diagnostics, and a complete fault-checked measurement cycle before the latch is released. If the retained run request is true, the BMS may then resume its normal startup sequence. Historical reasons remain available until reboot.

Configuration and initial TPL setup faults require a reboot. The ESP32, EMS, and Home Assistant cannot override a blocking STM32 fault.

5. Interfaces and ownership

Information or action	Owner	Consumer
Cell voltages and temperatures	STM32 BMS	Diagnostics, inverter limits, telemetry
Battery current and SoC	STM32 BMS	Inverter, EMS, Home Assistant
Battery safety limits	STM32 BMS	Inverter and EMS
Contactor and precharge	STM32 BMS	Physical HV path
Inverter operating state	Inverter	ESP32 / EMS / Home Assistant
Requested charge/ discharge power	EMS	Inverter-facing control interface
Dashboards and automation	Home Assistant	User and household services

The protocol between these layers should carry explicit validity and freshness information. A stale value must never be interpreted as a current permission to charge, discharge, or close the contactor.

6. Current project structure

Repository area	Contents
`Hardware/Master`	Master schematic, PCB, harnesses, and production data.
`Hardware/ModuleBoard`	Battery module-board schematic, PCB, and production data.
`Software/Master`	STM32G491 firmware and platform configuration.
`Software/Companion`	Current FlexBMS BMS maintenance UI with direct USB, Gateway, and portable Windows package outputs.

`Software/Gateway`	ESP32 UART v1, Wi-Fi provisioning/recovery, compiled Companion serving, Home Assistant MQTT discovery, optional RS485 energy meters, OTA, and transport of STM32-produced GoodWe CAN diagnostics.
`Documentation/protocol/uart-v1.md`	Canonical framed BMS protocol for Gateway UART and direct USB CDC, including test vectors.
`Simulations`	Electrical simulation files used during hardware development.
`Documentation`	This system overview and its Typst build setup.

The repository contains the operational STM32, ESP32 Gateway, and FlexBMS Companion implementations. Home Assistant integration is active and EMHASS is running in shadow mode. The site-specific Home Assistant/EMHASS configuration is deployed outside this firmware repository; automatic inverter actuation remains planned.

7. Open topics

This overview is intentionally incomplete. The next useful documentation and validation additions are:

- confirmed as-built module, NTC, and high-voltage topology;
- controlled GoodWe charge, derating, timeout, and recovery evidence;
- system-level operating, fault, recovery, and shutdown policy;
- the as-deployed Home Assistant templates, EMHASS inputs, and executor contract;
- fixed-horizon price/load/PV tail generation and eventual terminal-value handling;
- per-phase fuse protection and coordination with the EV charger and heat pump;
- a characterized GoodWe standby/ECO sequence and measured auxiliary power; and
- automatic-control fallback behavior, manual override, and user permissions.